

CSA0904 – JAVA PROGRAMMING

ASSESSMENT TOOL 1-Coding Challenge

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1. Library Book Management using Classes & Encapsulation

Code:

```
import java.util.Scanner;

class Book {

    private int bookId;
    private String title;
    private String author;
    private double price;
    private boolean issued;

    Book(int bookId, String title, String author, double price) {
        this.bookId = bookId;
        this.title = title;
        this.author = author;
        this.price = price;
        this.issued = false;
    }

    public int getBookId() {
        return bookId;
    }

    public void issueBook() {
        if (!issued) {
            issued = true;
            System.out.println("Book Issued Successfully.");
        }
    }
}
```

```

        } else {
            System.out.println("Book Already Issued.");
        }
    }

    public void returnBook() {
        if (issued) {
            issued = false;
            System.out.println("Book Returned Successfully.");
        } else {
            System.out.println("Book is Not Issued.");
        }
    }

    public void displayBook() {
        System.out.println("-----");
        System.out.println("Book ID : " + bookId);
        System.out.println("Title  : " + title);
        System.out.println("Author : " + author);
        System.out.println("Price  : " + price);
        System.out.println("Status : " + (issued ? "Issued" : "Available"));
        System.out.println("-----");
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Book books[] = new Book[10];
        int count = 0;
        int choice;
        do {

```

```
System.out.println("\n===== Library Book Management =====");
System.out.println("1. Add Book");
System.out.println("2. Display All Books");
System.out.println("3. Search Book");
System.out.println("4. Issue Book");
System.out.println("5. Return Book");
System.out.println("6. Exit");
System.out.print("Enter your choice: ");
choice = sc.nextInt();
switch (choice) {
    case 1:
        System.out.print("Enter Book ID: ");
        int id = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Title: ");
        String title = sc.nextLine();
        System.out.print("Enter Author: ");
        String author = sc.nextLine();
        System.out.print("Enter Price: ");
        double price = sc.nextDouble();
        books[count] = new Book(id, title, author, price);
        count++;
        System.out.println("Book Added Successfully.");
        break;
    case 2:
        if (count == 0) {
            System.out.println("No Books Available.");
        } else {
            for (int i = 0; i < count; i++) {
                books[i].displayBook();
            }
        }
    }
}
```

```

        }
    }
    break;
case 3:
    System.out.print("Enter Book ID to Search: ");
    id = sc.nextInt();
    boolean found = false;
    for (int i = 0; i < count; i++) {
        if (books[i].getBookId() == id) {
            books[i].displayBook();
            found = true;
            break;
        }
    }
    if (!found)
        System.out.println("Book Not Found.");
    break;
case 4:
    System.out.print("Enter Book ID to Issue: ");
    id = sc.nextInt();
    found = false;
    for (int i = 0; i < count; i++) {
        if (books[i].getBookId() == id) {
            books[i].issueBook();
            found = true;
            break;
        }
    }
    if (!found)
        System.out.println("Book Not Found.");

```

```
        break;
    case 5:
        System.out.print("Enter Book ID to Return: ");
        id = sc.nextInt();
        found = false;
        for (int i = 0; i < count; i++) {
            if (books[i].getBookId() == id) {
                books[i].returnBook();
                found = true;
                break;
            }
        }
        if (!found)
            System.out.println("Book Not Found.");
        break;
    case 6:
        System.out.println("Thank You...");
        break;
    default:
        System.out.println("Invalid Choice.");
    }
} while (choice != 6);
sc.close();
}
```

```
1- import java.util.Scanner;
2- class Book {
3-     private int bookId;
4-     private String title;
5-     private String author;
6-     private double price;
7-     private boolean issued;
8-     Book(int bookId, String title, String author, double price) {
9-         this.bookId = bookId;
10-        this.title = title;
11-        this.author = author;
12-        this.price = price;
13-        this.issued = false;
14-    }
15-     public int getBookId() {
16-         return bookId;
17-     }
18-     public void issueBook() {
19-         if (!issued) {
20-             issued = true;
21-             System.out.println("Book Issued Successfully.");
22-         } else {
23-             System.out.println("Book Already Issued.");
24-         }
25-     }
26-     public void returnBook() {
27-         if (issued) {
28-             issued = false;
29-             System.out.println("Book Returned Successfully.");
30-         } else {
31-             // This block is not visible in the screenshot but would logically follow
32-         }
33-     }
34- }
```

Output

```
===== Library Book Management =====
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter your choice: 1
Enter Book ID: 101
Enter Title: Java Programming
Enter Author: James gosling
Enter Price: 450
Book Added Successfully.

===== Library Book Management =====
1. Add Book
2. Display All Books
3. Search Book
4. Issue Book
5. Return Book
6. Exit
Enter your choice: 2

-----
Book ID : 101
Title   : Java Programming
Author  : James gosling
Price   : 450.0
Status  : Available
-----
```

2.Employee Payroll System using Inheritance

Code:

```
import java.util.Scanner;

class Employee {

    int empId;

    String name;

    Employee(int empId, String name) {

        this.empId = empId;

        this.name = name;

    }

    public double calculateSalary() {

        return 0;

    }

    public void display() {

        System.out.println("\nEmployee ID : " + empId);

        System.out.println("Employee Name : " + name);

        System.out.println("Salary : " + calculateSalary());

    }

}
```

```
}
```

```
class PermanentEmployee extends Employee {
```

```
    double basicSalary, hra, da;
```

```
    PermanentEmployee(int empId, String name, double basicSalary) {
```

```
        super(empId, name);
```

```
        this.basicSalary = basicSalary;
```

```
        hra = basicSalary * 0.20;
```

```
        da = basicSalary * 0.10;
```

```
    }
```

```
    @Override
```

```
    public double calculateSalary() {
```

```
        return basicSalary + hra + da;
```

```
    }
```

```
}
```

```
class ContractEmployee extends Employee {
```

```
    int hoursWorked;
```

```
    double ratePerHour;
```

```
    ContractEmployee(int empId, String name, int hoursWorked, double ratePerHour) {
```

```
        super(empId, name);
```

```
        this.hoursWorked = hoursWorked;
```

```
        this.ratePerHour = ratePerHour;
```

```
    }
```

```
    @Override
```

```
    public double calculateSalary() {
```

```

        return hoursWorked * ratePerHour;
    }
}

public class Main {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("1. Permanent Employee");
        System.out.println("2. Contract Employee");
        System.out.print("Enter Choice: ");
        int choice = sc.nextInt();

        Employee emp;

        if (choice == 1) {

            System.out.print("Enter Employee ID: ");
            int id = sc.nextInt();
            sc.nextLine();

            System.out.print("Enter Employee Name: ");
            String name = sc.nextLine();

            System.out.print("Enter Basic Salary: ");
            double salary = sc.nextDouble();

            emp = new PermanentEmployee(id, name, salary);

```



```

    } else {

        System.out.print("Enter Employee ID: ");

        int id = sc.nextInt();

        sc.nextLine();

        System.out.print("Enter Employee Name: ");

        String name = sc.nextLine();

        System.out.print("Enter Hours Worked: ");

        int hours = sc.nextInt();

        System.out.print("Enter Rate Per Hour: ");

        double rate = sc.nextDouble();

        emp = new ContractEmployee(id, name, hours, rate);

    }

    System.out.println("\n===== Employee Details =====");

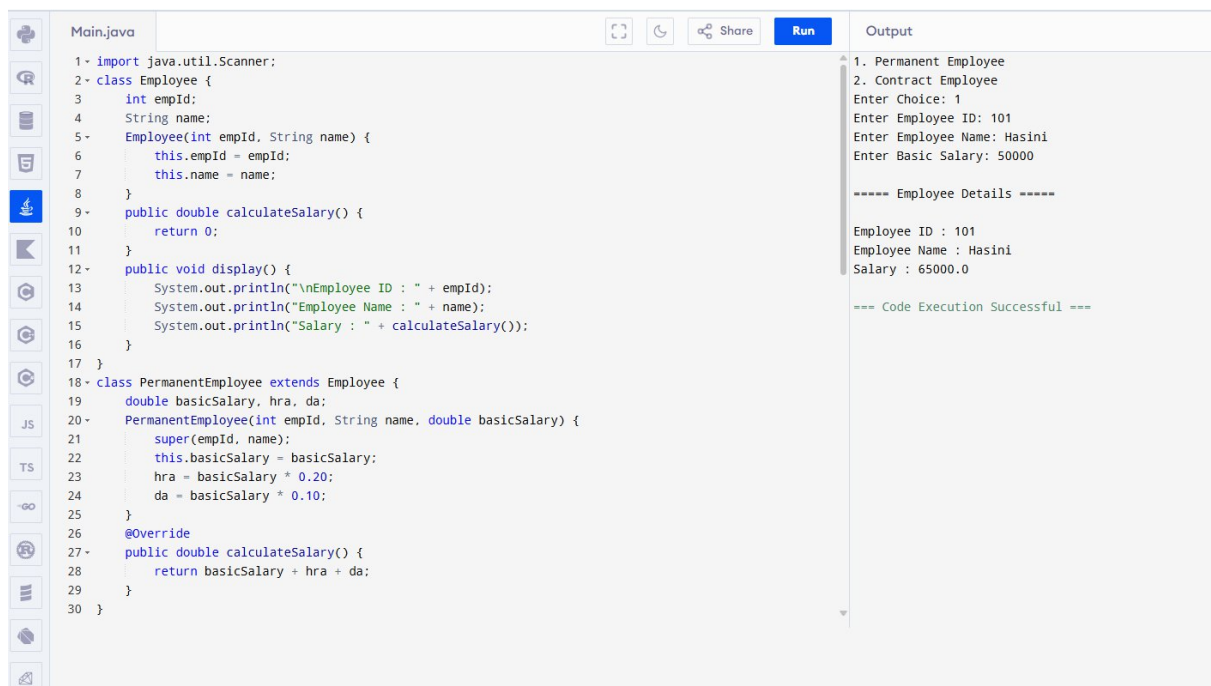
    emp.display();

    sc.close();

}

}

```



The screenshot shows a Java IDE with a file named 'Main.java'. The code defines an 'Employee' class with attributes 'empId' and 'name', and methods 'calculateSalary()' and 'display()'. It also defines a 'PermanentEmployee' class that extends 'Employee' and overrides 'calculateSalary()'. The output window shows the execution of the program, where a 'Contract Employee' is created with ID 101, name 'Hasini', and a basic salary of 50000. The final output shows the employee details: 'Employee ID : 101', 'Employee Name : Hasini', and 'Salary : 65000.0'.

```

Main.java
1- import java.util.Scanner;
2- class Employee {
3-     int empId;
4-     String name;
5-     Employee(int empId, String name) {
6-         this.empId = empId;
7-         this.name = name;
8-     }
9-     public double calculateSalary() {
10-         return 0;
11-     }
12-     public void display() {
13-         System.out.println("\nEmployee ID : " + empId);
14-         System.out.println("Employee Name : " + name);
15-         System.out.println("Salary : " + calculateSalary());
16-     }
17- }
18- class PermanentEmployee extends Employee {
19-     double basicSalary, hra, da;
20-     PermanentEmployee(int empId, String name, double basicSalary) {
21-         super(empId, name);
22-         this.basicSalary = basicSalary;
23-         hra = basicSalary * 0.20;
24-         da = basicSalary * 0.10;
25-     }
26-     @Override
27-     public double calculateSalary() {
28-         return basicSalary + hra + da;
29-     }
30- }

```

Output

```

1. Permanent Employee
2. Contract Employee
Enter Choice: 1
Enter Employee ID: 101
Enter Employee Name: Hasini
Enter Basic Salary: 50000

===== Employee Details =====

Employee ID : 101
Employee Name : Hasini
Salary : 65000.0

=== Code Execution Successful ===

```

3.Vehicle Rental System using Runtime Polymorphism

Code:

```
import java.util.Scanner;

class Vehicle {
    int vehicleId;
    String vehicleName;
    Vehicle(int vehicleId, String vehicleName) {
        this.vehicleId = vehicleId;
        this.vehicleName = vehicleName;
    }
    public double calculateRent(int days) {
        return 0;
    }
    public void display(int days) {
        System.out.println("\n===== Rental Bill =====");
        System.out.println("Vehicle ID   : " + vehicleId);
        System.out.println("Vehicle Name : " + vehicleName);
        System.out.println("Days       : " + days);
        System.out.println("Total Rent  : " + calculateRent(days));
    }
}

class Car extends Vehicle {
    Car(int id, String name) {
        super(id, name);
    }
    @Override
    public double calculateRent(int days) {
        return days * 1500;
    }
}
```

```
class Bike extends Vehicle {
    Bike(int id, String name) {
        super(id, name);
    }
    @Override
    public double calculateRent(int days) {
        return days * 500;
    }
}

class Bus extends Vehicle {
    Bus(int id, String name) {
        super(id, name);
    }
    @Override
    public double calculateRent(int days) {
        return days * 3000;
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Vehicle[] vehicles = new Vehicle[3];
        vehicles[0] = new Car(101, "Car");
        vehicles[1] = new Bike(102, "Bike");
        vehicles[2] = new Bus(103, "Bus");
        System.out.println("===== Vehicle Rental System =====");
        System.out.println("1. Car");
        System.out.println("2. Bike");
        System.out.println("3. Bus");
        System.out.print("Enter your choice: ");
    }
}
```

```

        int choice = sc.nextInt();

        System.out.print("Enter Number of Rental Days: ");

        int days = sc.nextInt();

        switch (choice) {

            case 1:

                vehicles[0].display(days);

                break;

            case 2:

                vehicles[1].display(days);

                break;

            case 3:

                vehicles[2].display(days);

                break;

            default:

                System.out.println("Invalid Choice");

        }

        sc.close();

    }

}

```

The screenshot shows an IDE with a file named 'Main.java'. The code defines a base class 'Vehicle' with attributes 'vehicleId' and 'vehicleName', and methods 'calculateRent' and 'display'. It also defines two subclasses, 'Car' and 'Bike', which inherit from 'Vehicle'. The 'Car' class overrides 'calculateRent' to return 'days * 1500'. The 'Bike' class is also defined but its implementation is not fully visible. The output window shows the execution of the program, displaying a menu of vehicle options (1. Car, 2. Bike, 3. Bus), the user's choice (2), the number of rental days (5), and a detailed rental bill for a Bike with ID 102, showing a total rent of 2500.0.

```

Main.java
1- import java.util.Scanner;
2- class Vehicle {
3-     int vehicleId;
4-     String vehicleName;
5-     Vehicle(int vehicleId, String vehicleName) {
6-         this.vehicleId = vehicleId;
7-         this.vehicleName = vehicleName;
8-     }
9-     public double calculateRent(int days) {
10-         return 0;
11-     }
12-     public void display(int days) {
13-         System.out.println("\n----- Rental Bill -----");
14-         System.out.println("Vehicle ID   : " + vehicleId);
15-         System.out.println("Vehicle Name : " + vehicleName);
16-         System.out.println("Days       : " + days);
17-         System.out.println("Total Rent  : " + calculateRent(days));
18-     }
19- }
20- class Car extends Vehicle {
21-     Car(int id, String name) {
22-         super(id, name);
23-     }
24-     @Override
25-     public double calculateRent(int days) {
26-         return days * 1500;
27-     }
28- }
29- class Bike extends Vehicle {
30-     Bike(int id, String name) {

```

```

Output
----- Vehicle Rental System -----
1. Car
2. Bike
3. Bus
Enter your choice: 2
Enter Number of Rental Days: 5

----- Rental Bill -----
Vehicle ID   : 102
Vehicle Name : Bike
Days       : 5
Total Rent  : 2500.0

=== Code Execution Successful ===

```

4. Bank Account System using Data Hiding

Code:

```
import java.util.Scanner;

class BankAccount {
    private int accountNumber;
    private String holderName;
    private double balance;

    // Constructor
    BankAccount(int accountNumber, String holderName, double balance) {
        this.accountNumber = accountNumber;
        this.holderName = holderName;
        this.balance = balance;
    }

    // Deposit Method
    public void deposit(double amount) {
        if (amount > 0) {
            balance += amount;
            System.out.println("Amount Deposited Successfully.");
        } else {
            System.out.println("Invalid Deposit Amount.");
        }
    }

    public void withdraw(double amount) {
        if (amount <= balance && amount > 0) {
            balance -= amount;
            System.out.println("Amount Withdrawn Successfully.");
        } else {
            System.out.println("Insufficient Balance or Invalid Amount.");
        }
    }
}
```

```

    public void checkBalance() {
        System.out.println("\n===== Account Details =====");
        System.out.println("Account Number : " + accountNumber);
        System.out.println("Holder Name   : " + holderName);
        System.out.println("Balance      : " + balance);
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Account Number: ");
        int accNo = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Account Holder Name: ");
        String name = sc.nextLine();
        System.out.print("Enter Initial Balance: ");
        double balance = sc.nextDouble();
        BankAccount account = new BankAccount(accNo, name, balance);
        int choice;
        do {
            System.out.println("\n===== Bank Account System =====");
            System.out.println("1. Deposit");
            System.out.println("2. Withdraw");
            System.out.println("3. Check Balance");
            System.out.println("4. Exit");
            System.out.print("Enter your choice: ");
            choice = sc.nextInt();
            switch (choice) {
                case 1:
                    System.out.print("Enter Deposit Amount: ");

```

```

        double deposit = sc.nextDouble();

        account.deposit(deposit);

        break;
    case 2:

        System.out.print("Enter Withdrawal Amount: ");

        double withdraw = sc.nextDouble();

        account.withdraw(withdraw);

        break;
    case 3:

        account.checkBalance();

        break;
    case 4:

        System.out.println("Thank You...");

        break;
    default:

        System.out.println("Invalid Choice.");

    }

    } while (choice != 4);

    sc.close();

}

}

```

The screenshot shows an IDE with a file named 'Main.java'. The code implements a Bank Account System with a constructor, deposit, and withdraw methods. The output window shows the program's execution, including account details, menu prompts, and successful transactions.

```

Main.java
1- import java.util.Scanner;
2
3- class BankAccount {
4-     private int accountNumber;
5-     private String holderName;
6-     private double balance;
7
8-     // Constructor
9-     BankAccount(int accountNumber, String holderName, double balance) {
10-         this.accountNumber = accountNumber;
11-         this.holderName = holderName;
12-         this.balance = balance;
13-     }
14
15-     // Deposit Method
16-     public void deposit(double amount) {
17-         if (amount > 0) {
18-             balance += amount;
19-             System.out.println("Amount Deposited Successfully.");
20-         } else {
21-             System.out.println("Invalid Deposit Amount.");
22-         }
23-     }
24
25-     // Withdraw Method
26-     public void withdraw(double amount) {
27-         if (amount <= balance && amount > 0) {
28-             balance -= amount;
29-             System.out.println("Amount Withdrawn Successfully.");
30-         } else {

```

```

Output
Enter Account Number: 1001
Enter Account Holder Name: Hasini
Enter Initial Balance: 5000

===== Bank Account System =====
1. Deposit
2. Withdraw
3. Check Balance
4. Exit
Enter your choice: 2
Enter Withdrawal Amount: 4000
Amount Withdrawn Successfully.

===== Bank Account System =====
1. Deposit
2. Withdraw
3. Check Balance
4. Exit
Enter your choice: 1
Enter Deposit Amount: 4000
Amount Deposited Successfully.

===== Bank Account System =====
1. Deposit
2. Withdraw
3. Check Balance
4. Exit
Enter your choice: 3

===== Account Details =====

```

5. University Admission System using Method Overloading

Code:

```
import java.util.Scanner;

class Admission {

    void calculateFee(double fee) {

        System.out.println("Undergraduate Fee : " + fee);

    }

    void calculateFee(double fee, int pg) {

        System.out.println("Postgraduate Fee : " + fee);

    }

    void calculateFee(double fee, double scholarship) {

        double finalFee = fee - scholarship;

        System.out.println("Scholarship Amount : " + scholarship);

        System.out.println("Final Fee : " + finalFee);

    }

}

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        Admission ad = new Admission();

        System.out.println("==== University Admission System =====");

        System.out.println("1. Undergraduate Student");

        System.out.println("2. Postgraduate Student");

        System.out.println("3. Scholarship Student");

        System.out.print("Enter your choice: ");

        int choice = sc.nextInt();

        switch (choice) {

            case 1:

                System.out.print("Enter Undergraduate Fee: ");

                double ugFee = sc.nextDouble();
```



```
ad.calculateFee(ugFee);
```

```
break;
```

case 2:

```
System.out.print("Enter Postgraduate Fee: ");
```

```
double pgFee = sc.nextDouble();
```

```
ad.calculateFee(pgFee, 1);
```

```
break;
```

case 3:

```
System.out.print("Enter Original Fee: ");
```

```
double fee = sc.nextDouble();
```

```
System.out.print("Enter Scholarship Amount: ");
```

```
double scholarship = sc.nextDouble();
```

```
ad.calculateFee(fee, scholarship);
```

```
break;
```

default:

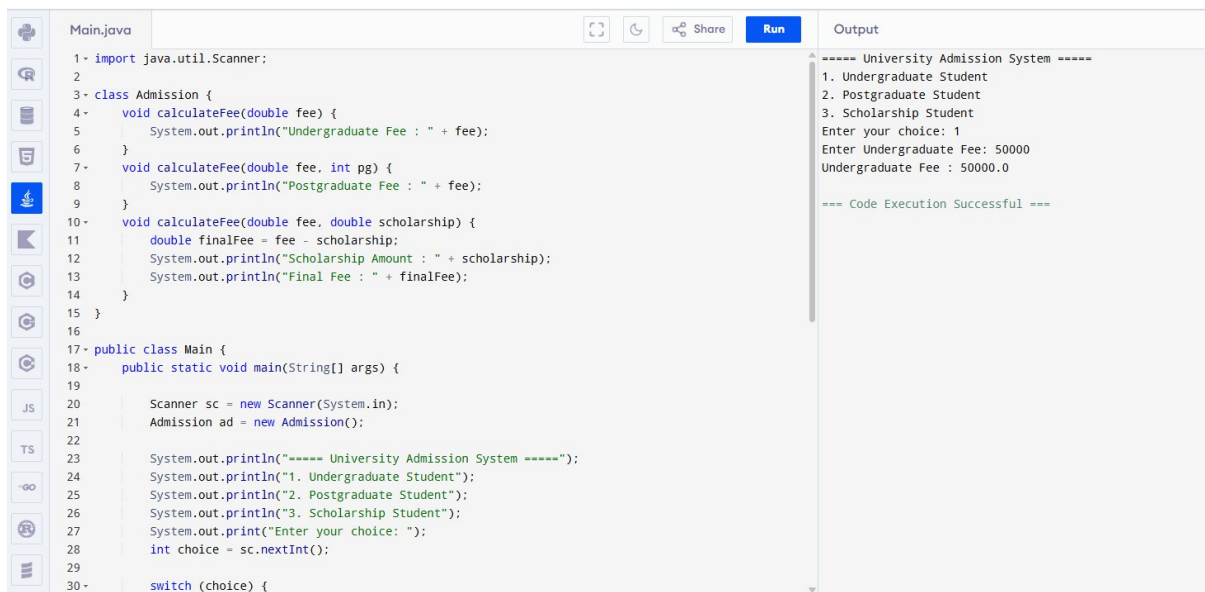
```
System.out.println("Invalid Choice");
```

```
}
```

```
sc.close();
```

```
}
```

```
}
```



The screenshot shows a Java IDE with a file named 'Main.java'. The code defines an 'Admission' class with three methods: 'calculateFee(double fee)', 'calculateFee(double fee, int pg)', and 'calculateFee(double fee, double scholarship)'. The 'Main' class has a 'main' method that prompts the user to choose between Undergraduate, Postgraduate, and Scholarship student options. It then calls the appropriate 'calculateFee' method based on the choice. The output window shows the execution of the program, displaying the menu, the user's choice of 1 (Undergraduate), and the resulting fee of 50000.0.

```
1- import java.util.Scanner;
2
3- class Admission {
4-     void calculateFee(double fee) {
5-         System.out.println("Undergraduate Fee : " + fee);
6-     }
7-     void calculateFee(double fee, int pg) {
8-         System.out.println("Postgraduate Fee : " + fee);
9-     }
10-    void calculateFee(double fee, double scholarship) {
11-        double finalFee = fee - scholarship;
12-        System.out.println("Scholarship Amount : " + scholarship);
13-        System.out.println("Final Fee : " + finalFee);
14-    }
15- }
16
17- public class Main {
18-     public static void main(String[] args) {
19-
20-         Scanner sc = new Scanner(System.in);
21-         Admission ad = new Admission();
22-
23-         System.out.println("==== University Admission System =====");
24-         System.out.println("1. Undergraduate Student");
25-         System.out.println("2. Postgraduate Student");
26-         System.out.println("3. Scholarship Student");
27-         System.out.print("Enter your choice: ");
28-         int choice = sc.nextInt();
29-
30-         switch (choice) {
```

Output

```
==== University Admission System =====
1. Undergraduate Student
2. Postgraduate Student
3. Scholarship Student
Enter your choice: 1
Enter Undergraduate Fee: 50000
Undergraduate Fee : 50000.0

=== Code Execution Successful ===
```

6. Shape Area Calculator using Abstract Class & Polymorphism

Code:

```
import java.util.Scanner;
```

```
abstract class Shape {
```

```
    abstract double area();
```

```
    void display() {
```

```
        System.out.println("Area = " + area());
```

```
    }
```

```
}
```

```
class Circle extends Shape {
```

```
    double radius;
```

```
    Circle(double radius) {
```

```
        this.radius = radius;
```

```
    }
```

```
    double area() {
```

```
        return 3.14 * radius * radius;
```

```
    }
```

```
}
```

```
class Rectangle extends Shape {
```

```
    double length, breadth;
```

```
    Rectangle(double length, double breadth) {
```

```
        this.length = length;
```

```
        this.breadth = breadth;
    }
```

```
double area() {
    return length * breadth;
}
}
```

```
class Triangle extends Shape {
    double base, height;
```

```
    Triangle(double base, double height) {
        this.base = base;
        this.height = height;
    }
```

```
    double area() {
        return 0.5 * base * height;
    }
}
```

```
public class Main {
    public static void main(String[] args) {
```

```
        Scanner sc = new Scanner(System.in);
```

```
        Shape[] shapes = new Shape[3];
```

```
        System.out.println("==== Shape Area Calculator =====");
```

```
        System.out.println("1. Circle");
```

```
System.out.println("2. Rectangle");
```

```
System.out.println("3. Triangle");
```

```
for (int i = 0; i < 3; i++) {
```

```
    System.out.print("\nEnter Shape Choice: ");
```

```
    int choice = sc.nextInt();
```

```
    switch (choice) {
```

```
        case 1:
```

```
            System.out.print("Enter Radius: ");
```

```
            double r = sc.nextDouble();
```

```
            shapes[i] = new Circle(r);
```

```
            break;
```

```
        case 2:
```

```
            System.out.print("Enter Length: ");
```

```
            double l = sc.nextDouble();
```

```
            System.out.print("Enter Breadth: ");
```

```
            double b = sc.nextDouble();
```

```
            shapes[i] = new Rectangle(l, b);
```

```
            break;
```

```
        case 3:
```

```
            System.out.print("Enter Base: ");
```

```
            double base = sc.nextDouble();
```

```
            System.out.print("Enter Height: ");
```

```
            double h = sc.nextDouble();
```

```
            shapes[i] = new Triangle(base, h);
```

```
break;
```

```
default:
```

```
System.out.println("Invalid Choice");
```

```
i--;
```

```
}
```

```
}
```

```
System.out.println("\n==== Area Details =====");
```

```
for (int i = 0; i < 3; i++) {
```

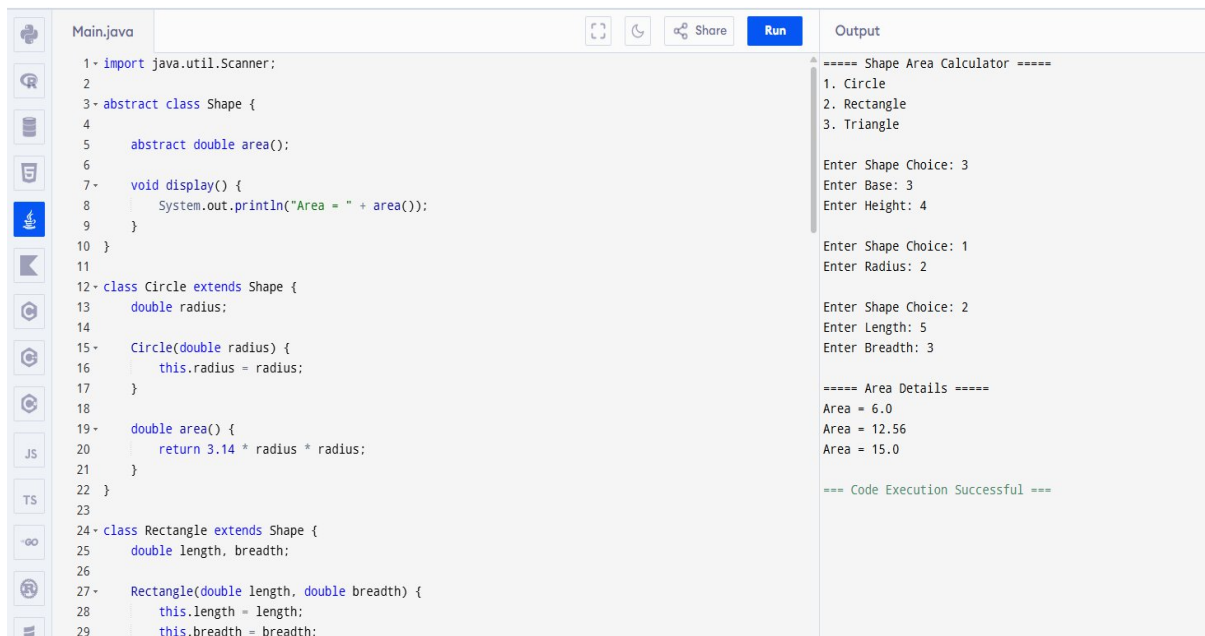
```
    shapes[i].display();
```

```
}
```

```
sc.close();
```

```
}
```

```
}
```



```
Main.java
1 import java.util.Scanner;
2
3 abstract class Shape {
4
5     abstract double area();
6
7     void display() {
8         System.out.println("Area = " + area());
9     }
10 }
11
12 class Circle extends Shape {
13     double radius;
14
15     Circle(double radius) {
16         this.radius = radius;
17     }
18
19     double area() {
20         return 3.14 * radius * radius;
21     }
22 }
23
24 class Rectangle extends Shape {
25     double length, breadth;
26
27     Rectangle(double length, double breadth) {
28         this.length = length;
29         this.breadth = breadth;
```

```
Output
===== Shape Area Calculator =====
1. Circle
2. Rectangle
3. Triangle

Enter Shape Choice: 3
Enter Base: 3
Enter Height: 4

Enter Shape Choice: 1
Enter Radius: 2

Enter Shape Choice: 2
Enter Length: 5
Enter Breadth: 3

===== Area Details =====
Area = 6.0
Area = 12.56
Area = 15.0

=== Code Execution Successful ===
```

7. Hospital Management System using Interfaces

Code:

```
import java.util.Scanner;

interface MedicalRecord {

    void addRecord();

    void displayRecord();

}

class Patient implements MedicalRecord {

    int patientId;

    String patientName;

    String disease;

    Scanner sc = new Scanner(System.in);

    public void addRecord() {

        System.out.print("Enter Patient ID: ");

        patientId = sc.nextInt();

        sc.nextLine();

        System.out.print("Enter Patient Name: ");

        patientName = sc.nextLine();

        System.out.print("Enter Disease: ");

        disease = sc.nextLine();

    }

    public void displayRecord() {

        System.out.println("\n----- Patient Record -----");

        System.out.println("Patient ID   : " + patientId);

        System.out.println("Patient Name : " + patientName);

        System.out.println("Disease     : " + disease);

    }

}
```

```

}

class Doctor implements MedicalRecord {
    int doctorId;
    String doctorName;
    String specialization;
    Scanner sc = new Scanner(System.in);
    public void addRecord() {
        System.out.print("Enter Doctor ID: ");
        doctorId = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Doctor Name: ");
        doctorName = sc.nextLine();
        System.out.print("Enter Specialization: ");
        specialization = sc.nextLine();
    }
    public void displayRecord() {
        System.out.println("\n----- Doctor Record -----");
        System.out.println("Doctor ID      : " + doctorId);
        System.out.println("Doctor Name   : " + doctorName);
        System.out.println("Specialization : " + specialization);
    }
}

```

```

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.println("===== Hospital Management System =====");
        System.out.println("1. Patient Record");
        System.out.println("2. Doctor Record");
        System.out.print("Enter your choice: ");
    }
}

```

```

int choice = sc.nextInt();

MedicalRecord record;

switch (choice) {

    case 1:

        record = new Patient();

        record.addRecord();

        record.displayRecord();

        break;

    case 2:

        record = new Doctor();

        record.addRecord();

        record.displayRecord();

        break;

    default:

        System.out.println("Invalid Choice");

}

sc.close();

}

}

```

```

Main.java
1- import java.util.Scanner;
2
3- interface MedicalRecord {
4     void addRecord();
5     void displayRecord();
6 }
7
8- class Patient implements MedicalRecord {
9     int patientId;
10    String patientName;
11    String disease;
12    Scanner sc = new Scanner(System.in);
13
14-    public void addRecord() {
15        System.out.print("Enter Patient ID: ");
16        patientId = sc.nextInt();
17        sc.nextLine();
18
19        System.out.print("Enter Patient Name: ");
20        patientName = sc.nextLine();
21
22        System.out.print("Enter Disease: ");
23        disease = sc.nextLine();
24    }
25
26-    public void displayRecord() {
27        System.out.println("\n----- Patient Record -----");
28        System.out.println("Patient ID : " + patientId);
29        System.out.println("Patient Name : " + patientName);
30        System.out.println("Disease : " + disease);

```

```

===== Hospital Management System =====
1. Patient Record
2. Doctor Record
Enter your choice: 1
Enter Patient ID: 101
Enter Patient Name: Bhavya
Enter Disease: fever

----- Patient Record -----
Patient ID : 101
Patient Name : Bhavya
Disease : fever

=== Code Execution Successful ===

```


8. Online Shopping Order System using Packages

Code:

```
import java.util.Scanner;

class Product {
    int productId;
    String productName;
    double price;

    Product(int productId, String productName, double price) {
        this.productId = productId;
        this.productName = productName;
        this.price = price;
    }
}

class Order {
    Product product;
    int quantity;

    Order(Product product, int quantity) {
        this.product = product;
        this.quantity = quantity;
    }

    double calculateTotal() {
        return product.price * quantity;
    }

    void displayBill() {
        System.out.println("\n===== Order Bill =====");
        System.out.println("Product : " + product.productName);
        System.out.println("Quantity : " + quantity);
        System.out.println("Price   : " + product.price);
        System.out.println("Total   : " + calculateTotal());
    }
}
```

```

}

public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.print("Enter Product ID: ");

        int id = sc.nextInt();

        sc.nextLine();

        System.out.print("Enter Product Name: ");

        String name = sc.nextLine();

        System.out.print("Enter Product Price: ");

        double price = sc.nextDouble();

        System.out.print("Enter Quantity: ");

        int quantity = sc.nextInt();

        Product p = new Product(id, name, price);

        Order o = new Order(p, quantity);

        o.displayBill();

        sc.close();

    }

}

```

The screenshot shows an IDE with a file named 'Main.java'. The code defines two classes: 'Product' and 'Order'. The 'Product' class has attributes 'productId', 'productName', and 'price', and a constructor. The 'Order' class has attributes 'product' and 'quantity', a constructor, a 'calculateTotal()' method, and a 'displayBill()' method. The 'Main' class contains the 'main' method, which uses a 'Scanner' to take user input for product ID, name, price, and quantity, then creates a 'Product' object, an 'Order' object, and calls 'displayBill()' to show the bill details.

```

1- import java.util.Scanner;
2
3- class Product {
4     int productId;
5     String productName;
6     double price;
7
8     Product(int productId, String productName, double price) {
9         this.productId = productId;
10        this.productName = productName;
11        this.price = price;
12    }
13 }
14
15- class Order {
16     Product product;
17     int quantity;
18
19     Order(Product product, int quantity) {
20         this.product = product;
21         this.quantity = quantity;
22     }
23
24     double calculateTotal() {
25         return product.price * quantity;
26     }
27
28     void displayBill() {
29         System.out.println("\n==== Order Bill =====");
30         System.out.print("Product : ");
31         System.out.print(product.productName);
32         System.out.print("\nQuantity : ");
33         System.out.print(quantity);
34         System.out.print("\nPrice : ");
35         System.out.print(product.price);
36         System.out.print("\nTotal : ");
37         System.out.print(calculateTotal());
38         System.out.println();
39     }
40 }

```

The output window shows the following text:

```

Enter Product ID: 121
Enter Product Name: laptop
Enter Product Price: 50000
Enter Quantity: 2

----- Order Bill -----
Product : laptop
Quantity : 2
Price : 50000.0
Total : 100000.0

--- Code Execution Successful ---

```

9.Student Result Analyzer using Arrays of Objects

Code:

```
import java.util.Scanner;

class Student {
    int rollNo;
    String name;
    int m1, m2, m3;
    int total;
    double average;
    char grade;

    void readData(Scanner sc) {
        System.out.print("Enter Roll Number: ");
        rollNo = sc.nextInt();
        sc.nextLine();
        System.out.print("Enter Name: ");
        name = sc.nextLine();
        System.out.print("Enter Marks in Subject 1: ");
        m1 = sc.nextInt();
        System.out.print("Enter Marks in Subject 2: ");
        m2 = sc.nextInt();
        System.out.print("Enter Marks in Subject 3: ");
        m3 = sc.nextInt();
    }

    void calculateResult() {
        total = m1 + m2 + m3;
        average = total / 3.0;
        if (average >= 90)
            grade = 'A';
    }
}
```

```

        else if (average >= 75)
            grade = 'B';
        else if (average >= 60)
            grade = 'C';
        else if (average >= 40)
            grade = 'D';
        else
            grade = 'F';
    }

    void display() {
        System.out.println("-----");
        System.out.println("Roll No : " + rollNo);
        System.out.println("Name   : " + name);
        System.out.println("Total   : " + total);
        System.out.println("Average : " + average);
        System.out.println("Grade   : " + grade);
    }
}

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Number of Students: ");
        int n = sc.nextInt();
        Student[] s = new Student[n];
        for (int i = 0; i < n; i++) {
            System.out.println("\nEnter Details of Student " + (i + 1));
            s[i] = new Student();
            s[i].readData(sc);
            s[i].calculateResult();
        }
    }
}

```

```

        System.out.println("\n===== Student Results =====");

        int topper = 0;

        for (int i = 0; i < n; i++) {

            s[i].display();

            if (s[i].total > s[topper].total) {

                topper = i;

            }

        }

        System.out.println("\n===== Class Topper =====");

        System.out.println("Roll No : " + s[topper].rollNo);

        System.out.println("Name   : " + s[topper].name);

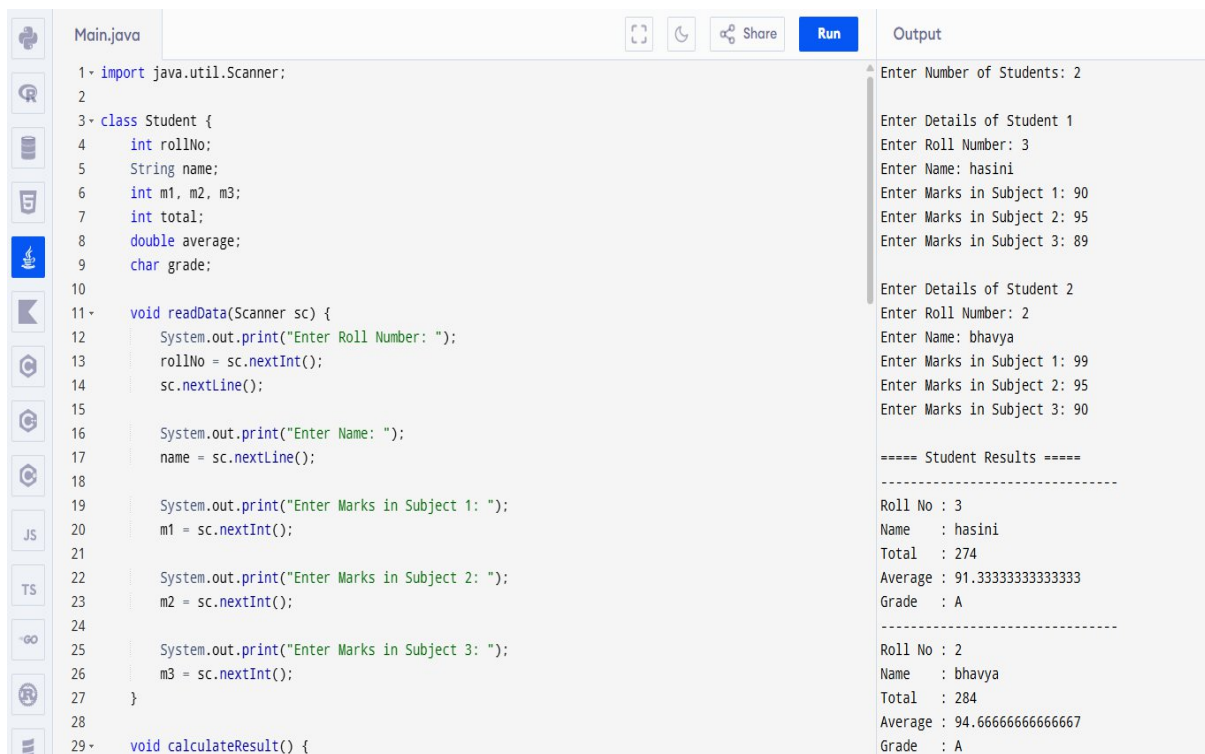
        System.out.println("Total   : " + s[topper].total);

        sc.close();

    }

}

```



The screenshot shows a Java IDE with a file named 'Main.java'. The code defines a 'Student' class with attributes for roll number, name, marks in three subjects, total, average, and grade. It includes methods to read student data and calculate results. The output window shows the execution of the program, where two students are entered: Student 1 (hasini) and Student 2 (bhavya). The results for Student 1 are displayed, showing a total of 274, an average of 91.33, and a grade of A.

```

Main.java
1 import java.util.Scanner;
2
3 class Student {
4     int rollNo;
5     String name;
6     int m1, m2, m3;
7     int total;
8     double average;
9     char grade;
10
11 void readData(Scanner sc) {
12     System.out.print("Enter Roll Number: ");
13     rollNo = sc.nextInt();
14     sc.nextLine();
15
16     System.out.print("Enter Name: ");
17     name = sc.nextLine();
18
19     System.out.print("Enter Marks in Subject 1: ");
20     m1 = sc.nextInt();
21
22     System.out.print("Enter Marks in Subject 2: ");
23     m2 = sc.nextInt();
24
25     System.out.print("Enter Marks in Subject 3: ");
26     m3 = sc.nextInt();
27 }
28
29 void calculateResult() {

```

Output

```

Enter Number of Students: 2

Enter Details of Student 1
Enter Roll Number: 3
Enter Name: hasini
Enter Marks in Subject 1: 90
Enter Marks in Subject 2: 95
Enter Marks in Subject 3: 89

Enter Details of Student 2
Enter Roll Number: 2
Enter Name: bhavya
Enter Marks in Subject 1: 99
Enter Marks in Subject 2: 95
Enter Marks in Subject 3: 90

===== Student Results =====
-----
Roll No : 3
Name   : hasini
Total   : 274
Average : 91.33333333333333
Grade   : A
-----
Roll No : 2
Name   : bhavya
Total   : 284
Average : 94.66666666666667
Grade   : A

```

10. Smart Home Device Controller using Hierarchical Inheritance

Code:

```
import java.util.Scanner;

class Device {
    String deviceName;

    Device(String deviceName) {
        this.deviceName = deviceName;
    }

    void turnOn() {
        System.out.println(deviceName + " is ON");
    }

    void turnOff() {
        System.out.println(deviceName + " is OFF");
    }

    double powerConsumption() {
        return 0;
    }

    void display() {
        turnOn();

        System.out.println("Power Consumption : " + powerConsumption() + " Watts");
        turnOff();
    }
}

class Light extends Device {
    Light() {
        super("Light");
    }

    @Override
    double powerConsumption() {
        return 20;
    }
}
```

```

    }
}
class Fan extends Device {
    Fan() {
        super("Fan");
    }
    @Override
    double powerConsumption() {
        return 75;
    }
}
class AirConditioner extends Device {
    AirConditioner() {
        super("Air Conditioner");
    }
    @Override
    double powerConsumption() {
        return 1500;
    }
}
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        Device device;
        System.out.println("==== Smart Home Device Controller =====");
        System.out.println("1. Light");
        System.out.println("2. Fan");
        System.out.println("3. Air Conditioner");
        System.out.print("Enter your choice: ");
    }
}

```

```

int choice = sc.nextInt();

switch (choice) {

    case 1:

        device = new Light();

        break;

    case 2:

        device = new Fan();

        break;

    case 3:

        device = new AirConditioner();

        break;

    default:

        System.out.println("Invalid Choice");

        sc.close();

        return;

}

System.out.println("\n===== Device Details =====");

device.display();

sc.close();

}

}

```

```

Main.java
1- import java.util.Scanner;
2
3- class Device {
4     String deviceName;
5
6     Device(String deviceName) {
7         this.deviceName = deviceName;
8     }
9
10    void turnOn() {
11        System.out.println(deviceName + " is ON");
12    }
13
14    void turnOff() {
15        System.out.println(deviceName + " is OFF");
16    }
17
18    double powerConsumption() {
19        return 0;
20    }
21
22    void display() {
23        turnOn();
24        System.out.println("Power Consumption : " + powerConsumption() + " Watts");
25        turnOff();
26    }
27 }
28
29 class Light extends Device {
30

```

```

Output
===== Smart Home Device Controller =====
1. Light
2. Fan
3. Air Conditioner
Enter your choice: 2

===== Device Details =====
Fan is ON
Power Consumption : 75.0 Watts
Fan is OFF

=== Code Execution Successful ===

```